Conditional statement OR Control statement

- they were used when we want a piece of code to be executed if a particular condition is to true or false.
- if the key will used to write a contional statement.

DIFFERNET TYPES OF CONDITIONAL STATEMENTS:

- if statement
- if else statement
- multiple statement
- if...elif....else statement
- nested if statement

:INDENTATION

- suite----> A group of statement written in python together.
  - indentation refers to a space that are used in the beginning of the statement, and the statement with same indentation belongs to same suite.
  - indentation refers to 4 spaces , but it increased or decreased by programmers.

IF STATEMENT :

- syntax

```
if condition:
   code_1
   code_2
```

- when the if condition are true then your code will get executed.>

In [12]:
```
my_str = input("enter a yes or no: ")
if my_str == "yes":
    print("YES")
    print('This is what i said')
    print('Your response is good')
```

```
enter a yes or no: yes
YES
This is what i said
Your response is good
```

In [14]:
```
my_str = input("enter a yes or no: ")
if my_str == "yes":
    print("YES")
    print('This is what i said')
print('Your response is good')
```

```
enter a yes or no: no
Your response is good
```

In [18]:
```
a = int(input("enter a number: "))
if a % 2 == 0: # to check the whether the number is even or not
    print(f"{a} number is EVEN")

print('GOOD DAY!')
```

```
enter a number: 26
26 number is EVEN
GOOD DAY!
```

In [19]:
```
a = int(input("enter a number: "))
if a % 2 == 0: # to check the whether the number is even or not
    print(f"{a} number is EVEN")

print('GOOD DAY!')
```

```
enter a number: 11
GOOD DAY!
```

IF ELSE STATEMENT :

- Syntax

```
if condition:
   code_1

else:
   code_2
```

- When the if condition true the condition inside IF will executed else will ELSE condition executed.

In [20]:
```
a = int(input('enter a number: '))

if a % 2 == 0:
    print(f'{a} is an EVEN number')
else:
    print(f'{a} is an odd number')
```

```
enter a number: 5
5 is an odd number
```

In [21]:
```
a = int(input('enter a number: '))

if a % 2 == 0:
    print(f'{a} is an EVEN number')
else:
    print(f'{a} is an odd number')
```

```
enter a number: 22
22 is an EVEN number
```

MULTIPLE IF STATEMENT :

- Syntax

```
- if condition_1:
      code

- if condition_2:
      code

- if condition_3:
      code
    .
    .
    .
    .
  if condition_n:
      code
```

In [26]:
```
a = eval(input('enter a number: '))

if a == 1:
    print('one')
if a == 2:
    print('Two')
if a == 3:
    print('three')

print('GOOD DAY!!')
```

```
enter a number: 3
three
GOOD DAY!!
```

In [27]:
```
a = eval(input('enter a number: '))

if a % 2 == 0:
    print(f'{a} is divisible by 2')
if a % 3 == 0:
    print(f'{a} is divisible by 3')
if a % 5 == 0:
    print(f'{a} is divisible by 5')

print('GOOD DAY!!')
```

```
enter a number: 10
10 is divisible by 2
10 is divisible by 5
GOOD DAY!!
```

In [4]:
```
a = eval(input('enter a number: '))

import time  # PROGRAM EXECUTION TIMe
ss= time.time()
print(ss)


if a % 2 == 0:
    print(f'{a} is divisible by 2')
if a % 3 == 0:
    print(f'{a} is divisible by 3')
if a % 5 == 0:
    print(f'{a} is divisible by 5')

print('GOOD DAY!!')

end = time.time()
print(end)

delta = end - ss
print(delta)
```

```
enter a number: 30
1701250057.5491195
30 is divisible by 2
30 is divisible by 3
30 is divisible by 5
GOOD DAY!!
1701250057.550119
0.00099945068359375
```

Q - write a program to tell me that a given number is divisble by 2, 3 or 5

In [7]:
```
a = eval(input('enter a number: '))

import time  # PROGRAM EXECUTION TIME
ss= time.time()
print(ss)


if a % 2 == 0 and a % 3 != 0 and a % 5 != 0:
    print(f'{a} is divisible by 2')

if a % 2 != 0 and a % 3 == 0 and a % 5 != 0:
    print(f'{a} is divisible by 3')

if a % 2 != 0 and a % 3 != 0 and a % 5 == 0:
    print(f'{a} is divisible by 5')

if a % 2 == 0 and a % 3 != 0 and a % 5 == 0:
    print(f'{a} is divisible by 2 and 5')

if a % 2 == 0 and a % 3 == 0 and a % 5 != 0:
    print(f'{a} is divisible by 2 and 3')

if a % 2 != 0 and a % 3 == 0 and a % 5 == 0:
    print(f'{a} is divisible by 3 and 5')

if a % 2 == 0 and a % 3 == 0 and a % 5 == 0:
    print(f'{a} is divisible by 2,3 and 5')

print('GOOD DAY!!')

end = time.time()
print(end)

delta = end - ss
print(delta)
```

```
enter a number: 10
1701250529.233179
10 is divisible by 2 and 5
GOOD DAY!!
1701250529.2351758
0.001996755599975586
```

IF-ELIF-ELSE STATEMENT :

- Syntax

```
- if condition:
    code
```

In [4]:
```
#take 3 numbers from the user and going to check the maximum among the three


a = eval(input('enter a value : '))
b = eval(input('enter a value : '))
c = eval(input('enter a value : '))

if a >b and a>c:
    print(f'{a} is the greater num ')

elif b >a and b>c:
    print(f'{b} is the greater num ')

else:
    print(f'{c} is the greater num ')
```

```
enter a value : 12
enter a value : 21
enter a value : 2
21 is the greater num
```

In [5]:
```
#take 4 numbers from the user and going to check the maximum among the four

a = eval(input('enter a FIRST NUMBER : '))
b = eval(input('enter a SECOND NUMBER : '))
c = eval(input('enter a THIRD NUMBER : '))
d = eval(input('enter a FOURTH NUMBER: '))

if a >b and a>c and a>d:
    print(f'{a} is the greater num ')

elif b >a and b>c and b>d:
    print(f'{b} is the greater num ')

elif c>a and c>b and c>d:
    print(f'{c} is the greater num')

else:
    print(f'{d} is the greater num ')
```

```
enter a FIRST NUMBER : 12
enter a SECOND NUMBER : 32
enter a THIRD NUMBER : 45
enter a FOURTH NUMBER: 51
51 is the greater num
```

NESTED..IF STATEMENT :

- Syntax

```
- if condition:
        code1
    - if condition:
            code3
        - if condition:
                code5
        - else:
                code6
    - else:
          code4
```

if a student score 90 or more in all the three subjects(math, science and english) then the student has Well done!!! ,

if a student score <90 marks in any subject then it should try again

In [10]:
```
MATHS =   eval(input('enter the marks: '))
SCIENCE =eval(input('enter the marks: '))
ENGLISH = eval(input('enter the marks: '))

if MATHS>= 90:
    print(f'{MATHS} YOU SCORE GOOD!!')
    if SCIENCE >= 60:
        print(f'{SCIENCE} YOU SCORE GOOD!!')
        if ENGLISH >=70:
            print(f'{ENGLISH} YOU SCORE ENGLISH!!')
        else:
            print('WORK HARD IN ENGLISH')
    else:
        print('WORK HARD IN SCIENCE')
else:
    print('WORK HARD IN MATHS')
```

```
enter the marks: 95
enter the marks: 34
enter the marks: 44
95 YOU SCORE GOOD!!
WORK HARD IN SCIENCE
```

In [1]:
```
MATHS =   eval(input('enter the marks: '))
SCIENCE =eval(input('enter the marks: '))
ENGLISH = eval(input('enter the marks: '))

if MATHS>= 90:
    if SCIENCE >= 90:
        if ENGLISH >=90:
            print('WELL DONE !!')
        else:
            print('WORK HARD IN ENGLISH')
    else:
        print('WORK HARD IN SCIENCE')
else:
    print('WORK HARD IN MATHS')
```

```
enter the marks: 98
enter the marks: 91
enter the marks: 87
WORK HARD IN ENGLISH
```

In [6]:
```
# Take an alphabet from the user and check whetehr the alphabet is a vowel or c

vowel = 'aeiouAEIOU'
a = input('Enter an alphabet: ')

if a.isalpha():
    if a in vowel:
        print(f'{a} is an vowel')
    else:
        print(f'{a} is an consonant')
else:
    print(f'{a} is an alphabet')
```

```
Enter an alphabet: A
A is an vowel
```

In [3]:
```
alphanum1 = '123445abcder345'
a = input('Enter an alphabet or number: ')

if a.isalnum():
    if a in alphanum1:
        print(f'{a} is an alphanum')
    else:
        print(f'{a} is not alphanum')
else:
    print(f'{a} String is invalid!')
```

```
Enter an alphabet or number: $
$ String is invalid!
```

In [10]:
```
a = input('Enter an Number: ')

if a.isnumeric():
    if int(a) % 2 == 0:
        print(f'{a} is an EVEN')
    else:
        print(f'{a} is ODD')
else:
    print("INVALID INPUT! ENTER AN VALID NUMBER")
```

```
Enter an Number: ' '
INVALID INPUT! ENTER AN VALID NUMBER
```

LOOPS :

- we use loops for iterating a iterable again and again.
- Loops is an concept we use to execute code repeatedly.

TYPES OF LOOPS :

- For Loop
- While Loop

FOR LOOP :

- It always applicable on iterable obj(STRINGS, LISTS, TUPLES, DICTIONARY,SETS,FROZENSETS)
- Syntax:

```
- for var in iterable :
    code 1
    code 2
```

- where, var is an placeholder {PLACEHOLDER IS WHO HOLDS THE ELEMENT OF ITERABLE FOR SPECIFIC PERIOD OF TIME}

Iteration :

- Tranversing upon an element of iterable from the beginning to end is known as iteration.
- The number of iteration is equal to number of elements inside iterable object.

In [15]:
```
my_str = 'Python by Learnbay'

for i in my_str:
    print(i)
```

```
P
y
t
h
o
n

b
y

L
e
a
r
n
b
a
y
```

In [16]:
```
lst = [1,2,3,4,5,6,7]

for i in lst:
    print(i)
```

```
1
2
3
4
5
6
7
```

In [17]:
```
tupl = (1,2,3,4,5,6,7)

for i in tupl:
    print(i)
```

```
1
2
3
4
5
6
7
```

In [18]:
```
dictionary = {'name': 'Lavish', 'Age':21, 'City':'Kanpur'}

for i in dictionary:
    print(i)
```

```
name
Age
City
```

In [21]:
```
dictionary = {'name': 'Lavish', 'Age':21, 'City':'Kanpur'}

for i in dictionary.keys():
    print(i)
```

```
name
Age
City
```

In [19]:
```
dictionary = {'name': 'Lavish', 'Age':21, 'City':'Kanpur'}

for i in dictionary.values():
    print(i)
```

```
Lavish
21
Kanpur
```

In [20]:
```
dictionary = {'name': 'Lavish', 'Age':21, 'City':'Kanpur'}

for i in dictionary.items():
    print(i)
```

```
('name', 'Lavish')
('Age', 21)
('City', 'Kanpur')
```

In [22]:
```
set1 = {12,1,3,1,4,1,5,1,6,1,7}  # ---> DUPLICATES HAVE BEEN REMOVED

for i in set1:
    print(i)
```

```
1
3
4
5
6
7
12
```

In [28]:
```
set1 = {12,1,3,1,4,1,5,1,6,1,7}  # ---> DUPLICATES HAVE BEEN REMOVED
a = frozenset(set1)
print(a)

for i in a:
    print(i)
```

```
frozenset({1, 3, 4, 5, 6, 7, 12})
1
3
4
5
6
7
12
```

FOR LOOP : it does not work numeric values{ Int, Float, complex}

In [29]:
```
int1 = 23

for i in int1:
    print(i)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[29], line 3
      1 int1 = 23
----> 3 for i in int1:
      4     print(i)

TypeError: 'int' object is not iterable
```

In [30]:
```
flt2 = 24.55

for i in flt:
    print(i)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[30], line 3
      1 flt2 = 24.55
----> 3 for i in flt:
      4     print(i)

NameError: name 'flt' is not defined
```

In [31]:
```
compl = 2+3j

for i in compl:
    print(i)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[31], line 3
      1 compl = 2+3j
----> 3 for i in compl:
      4     print(i)

TypeError: 'complex' object is not iterable
```

In [36]:
```
# Cummulative sum:

sum = 0

for i in range(1,11):
    sum += i    # sum = sum + i
    print(sum)

print(f'{sum} is an total sum')
```

```
1
3
6
10
15
21
28
36
45
55
55 is an total sum
```

In [39]:
```
# FOR FINDING INDEX NUM?

my_str = 'Python Class'

for i in range(len(my_str)):
    print(i , my_str[i])
```

```
0 P
1 y
2 t
3 h
4 o
5 n
6 
7 C
8 l
9 a
10 s
11 s
```

In [43]:
```
# ENUMERATE : it is an function which used to mimic the code.

for i,j in enumerate(my_str):
    print(i,j)
```

```
0 P
1 y
2 t
3 h
4 o
5 n
6 
7 C
8 l
9 a
10 s
11 s
```

In [45]:
```
# LENGTH:

count = 0

for i in range(len(my_str)):
    count +=1 # count = count + i

print(count)
```

```
12
```

In [46]:
```
for i in range(5,-1,-1):
    print(i, my_str[i])
```

```
5 n
4 o
3 h
2 t
1 y
0 P
```

In [47]:
```
str1 = 'Python'

for i in range(-1 ,-len(str1)-1,-1):
    print(i,str1[i])
```

```
-1 n
-2 o
-3 h
-4 t
-5 y
-6 P
```

In [56]:
```
for i in range(-6,0,1):
    print(i,str1[i])
```

```
-6 P
-5 y
-4 t
-3 h
-2 o
-1 n
```

In [59]:
```
#Print every alternate elements inside a list using loops
lst = [10,20,30,40,50,60,70,80]

for i in range(0,len(lst),2):
    print(i,lst[i])
```

```
0 10
2 30
4 50
6 70
```

In [60]:
```
for i in range(-1,-len(lst),-2):
    print(i,lst[i])
```

```
-1 80
-3 60
-5 40
-7 20
```

In [65]:
```
my_str = "Peter piper picked a pack of pickeled peppers"

# COUNT ALL THE OCCURANCES OF "p"

count = 0

for i in my_str:
    if i == 'P' or i == 'p':
        count += 1 # count = count + 1

print(count)
```

```
9
```

In [67]:
```
# print all the index for all the occurance of the P

for i in range(0,len(my_str)):
    if my_str[i] == 'P' or my_str[i] == 'p':
        print(i)
```

```
0
6
8
12
21
29
38
40
41
```

In [69]:
```
# Print all the characters that are available at the even inde
str2 = 'My Name is Lavish Gangwani'


for i in range(0,len(str2),2):
    print(i, str2[i])
```

```
0 M
2
4 a
6 e
8 i
10
12 a
14 i
16 h
18 G
20 n
22 w
24 n
```

In [71]:
```
for i in range(0,len(str2)):
    if i % 2 == 0:
        print(i, str2[i])
```

```
0 M
2
4 a
6 e
8 i
10
12 a
14 i
16 h
18 G
20 n
22 w
24 n
```

In [2]:
```
rows = []
columns = []

for i in range(4):
    for j in range(4):
        result = i * j
        columns.append(result)
    rows.append(columns)
    columns=[]

rows
```

Out[2]:
```
[[0, 0, 0, 0], [0, 1, 2, 3], [0, 2, 4, 6], [0, 3, 6, 9]]
```

WHILE LOOP :

- While loops are used when we don't know the number of iteration or we want to execute loop infinite times.
- While loop can be used without iterables.

COMPONENTS OF WHILE LOOPS :

- Counter initialisation
- Condition
- Increment OR Decrement

-Syntax :

In [5]:
```
# We want to print numbers from 0 to 10

i = 0

while i <= 10:
    print(i)
    i += 1        #increment operator
```

```
0
1
2
3
4
5
6
7
8
9
10
```

In [6]:
```
# We want to print numbers from 10 to 0

i = 10

while i >= 0:
    print(i)
    i -= 1     #decrement operator
```

```
10
9
8
7
6
5
4
3
2
1
0
```

In [8]:
```
my_str = 'Python by Learnbay'

start = 17
stop = 0

while start >= stop:
    print(start , my_str[start])
    start -= 1
```

```
17 y
16 a
15 b
14 n
13 r
12 a
11 e
10 L
9
8 y
7 b
6
5 n
4 o
3 h
2 t
1 y
0 P
```

In [10]:
```
str1 = 'Lavish Gangwani'

start = 0
stp = 14

while start <= stp:
    print(start, str1[start])
    start += 1
```

```
0 L
1 a
2 v
3 i
4 s
5 h
6
7 G
8 a
9 n
10 g
11 w
12 a
13 n
14 i
```

In [14]:
```
# W.A.P reversed a number by using while loop.

a = 2211

revrsed1 = 0

while a > 0:
    num = a % 10
    revrsed1 = revrsed1 * 10 + num
    a = a // 10

print(revrsed1)
```

```
1122
```

In [1]:
```
a = 3342

rev = 0

while a > 0:
    num = a % 10
    rev = rev * 10 + num
    a = a // 10

print(rev)
```

```
2433
```

While True :

- While true is always true so its executes code infinite times until we manage to terminate it by giving BREAK(TRANSFER STATEMENTS).
- BREAK ---> its a transfer statment use to terminate while true loops.

In [3]:
```
user_name = 'lavishgw22'

name = input('enter a user_name: ')

while True :
    if name == user_name:
        print(f'Hello.! {user_name}')
        print('You have successfully logged in!')
        break

    else:
        print('Invalid name?')
        name = input('enter a username: ')
```

```
enter a user_name: heleo2
Invalid name?
enter a username: mynameislavish
Invalid name?
enter a username: lavishboie
Invalid name?
enter a username: lavishgw22
Hello.! lavishgw22
You have successfully logged in!
```

In [7]:
```
a = 0

while a <=100:
    if a == 30:
        break

    else:
        print(a)
        a += 1
```

```
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
```

Nested loops :

- loop inside the loop are known as nested loops.
- when inner loops completes all iteration then outer loops iterate next elements.

```
- Syntax:
```

In [9]:
```
for i in range(4):
    for j in range(3):
        print(f'The value of i is {i} and the value of j is {j}')
        print('Hello')
    print('--------------------')
```

```
The value of i is 0 and the value of j is 0
Hello
The value of i is 0 and the value of j is 1
Hello
The value of i is 0 and the value of j is 2
Hello
--------------------
The value of i is 1 and the value of j is 0
Hello
The value of i is 1 and the value of j is 1
Hello
The value of i is 1 and the value of j is 2
Hello
--------------------
The value of i is 2 and the value of j is 0
Hello
The value of i is 2 and the value of j is 1
Hello
The value of i is 2 and the value of j is 2
Hello
--------------------
The value of i is 3 and the value of j is 0
Hello
The value of i is 3 and the value of j is 1
Hello
The value of i is 3 and the value of j is 2
Hello
--------------------
```

Enumerate :

- It takes an iterable an argument.
- It returns enumerate object.
- To fetch the value from enumerator we need to pass value in list([....]),a tuple ((...)) or a dictionary ({...}
- It maps the element of iterable one by one fdrom 0 to len(iterable)-1

In [1]:
```
a = 'Python'
e = enumerate(a)
print(e,type(e))
```

```
<enumerate object at 0x000002877C9EA930> <class 'enumerate'>
```

In [2]:
```
list(e)
```

Out[2]:
```
[(0, 'P'), (1, 'y'), (2, 't'), (3, 'h'), (4, 'o'), (5, 'n')]
```

In [3]:
```
tuple(e)
```

Out[3]:
```
()
```

In [4]:
```
a = 'Python'
e = enumerate(a)
print(e,type(e))
```

```
<enumerate object at 0x000002877D744A40> <class 'enumerate'>
```

In [5]:
```
dict(e)
```

Out[5]:
```
{0: 'P', 1: 'y', 2: 't', 3: 'h', 4: 'o', 5: 'n'}
```

In [6]:
```
a = 'Python'
e = enumerate(a)
print(e,type(e))
```

```
<enumerate object at 0x000002877D2FF6A0> <class 'enumerate'>
```

In [8]:
```
for i,j in enumerate(a):
    print(i,j)
```

```
0 P
1 y
2 t
3 h
4 o
5 n
```

In [10]:
```
aplhabet= {chr(i):i for i in range(ord('A'),ord('Z')+1)}
print(aplhabet)
```

```
{'A': 65, 'B': 66, 'C': 67, 'D': 68, 'E': 69, 'F': 70, 'G': 71, 'H': 72, 'I':
73, 'J': 74, 'K': 75, 'L': 76, 'M': 77, 'N': 78, 'O': 79, 'P': 80, 'Q': 81,
'R': 82, 'S': 83, 'T': 84, 'U': 85, 'V': 86, 'W': 87, 'X': 88, 'Y': 89, 'Z':
90}
```

In [12]:
```
a = {i:ord(i) for i in ('ABCDEFGHIJKLMNOPQRSTUVWXYZ')}
print(a)
```

```
{'A': 65, 'B': 66, 'C': 67, 'D': 68, 'E': 69, 'F': 70, 'G': 71, 'H': 72, 'I':
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